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Root responses to nitrogen pulse frequency under different nitrogen amounts

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ABSTRACT

Responses of morphology and biomass allocation of roots to frequency of nitrogen (N) pulse potentially influence the fitness of plants, but such responses may be determined by root size. We grew 12 plant species of three functional groups (grasses, forbs, and legumes) under two N pulse frequencies (high *vs.* low supply frequency) and two N amounts (high *vs.* low supply amount). Compared to low-amount N supply, high-amount N supply stimulated biomass accumulation and root growth by either increasing the thickness and length of roots or decreasing the root mass fraction. Compared to low-frequency N supply, high-frequency N supply improved biomass accumulation and root growth in forbs or grasses, but not in legumes. Furthermore, the magnitude of the response to N frequency was significantly negatively correlated with root size at the species scale, but this was only true when the N amount was high. We conclude that root responses to N frequency are related to plant functional types, and non-legume species is more sensitive to N frequency than legume species. Our results also suggest that root size is a determinant of root responses to N frequency when N supply amount is high.
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1. Introduction

Nitrogen (N) is a key nutrient element for plant growth (Elser et al., 2007; Vitousek and Howarth, 1991). As the root is commonly the most important organ by which plants acquire N from soils (Hodge, 2009; Hodge et al., 2009; Johnson and Biondini, 2001), its morphological plasticity and biomass allocation are important for plants to adapt to changing soil N concentration. When soil N is limited, plants may maximize N uptake efficiency by increasing the investment in root mass and/or modifying root morphology by, for example, enhancing lateral root length and root surface area (Johnson and Biondini, 2001; Mou et al., 2013; Robinson and Rorison, 1988). When soil N is abundant, plants may optimize N utilization efficiency by increasing leaf production or nitrogen concentration per leaf area (Bloom et al., 1985; Thornley, 1972).

There is increasing evidence that the response of roots to soil N is constrained by the inherent root structure and function of different species (Campbell et al., 1991; Cantarel et al., 2015; Fransen et al., 1998; Johnson and Biondini, 2001; Song et al., 2015; Wijesinghe et al., 2001; Yu et al., 2015). Species with large root systems have been reported to be more sensitive to temporal or spatial variation in N supply and possibly adjust root morphology and allocation more rapidly than do species with smaller root systems (Fransen et al., 1998; Johnson and Biondini, 2001). For instance, species with large root size (e.g., *Anthoxanthum odoratum*, *Holcus lanatus*, *Lolium perenne*) increased root mass in the high than in the low nutrient patches in a heterogeneous soil environment, but species with a smaller root size (e.g., *Festuca rubra*) did not (Fransen et al., 1998). Furthermore, morphological plasticity related to N availability is not only species-specific, but also varies among plant functional groups. For instance, plant species and functional groups often diverge in their N chemical form preference in response to fertilization (e.g., legumes sometimes show an apparent preference of nitrate over ammonium in an alpine meadow system; Song et al., 2015), which may influence the uptake efficiency of resources among functional groups, and also their morphological plasticity in response to changing in N availability. Besides, legumes often have the weaker growth vigor and morphological plasticity in response to N availability than grasses or forbs, especially when they fail to establish a plant-rhizobia mutuality during the seedling development period (Craine et al., 2003; Hodge et al., 2009). Therefore, comparing

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morphological plasticity between species and functional groups is necessary for a better understanding of the adaptive strategies of plants in response to changing N availability.

N loading is often considered as a discrete event in natural habitats, especially following unpredictable disturbance such as fire, rainfall and fertilization (Esque et al., 2010; Ivans et al., 2003). N loading of different frequencies increases the challenge for plants to acquire resources, but to some extent also provides the opportunity for plants to acclimatize to resource fluctuation (Bilbrough and Caldwell, 1997; Osone et al., 2014). One plausible prediction is that if high-frequency N supply may continually stabilize N availability in soils compared to low-frequency N supply, high-frequency N supply will improve plant performance. Meanwhile, plants with larger root systems may also exhibit a better adaptation to high-frequency N supply, because they are often larger in whole body size, are supported by a larger amount of photosynthates and N, and require more flexible root morphology in response to pulsing N supply (e.g., James and Richards, 2006). Another prediction is that the effect of N pulse frequency on plant performance may be modified by the amount of N supply. For instance, *Alternanthera sessilis* tended to increase leaf production and biomass accumulation under high-frequency than under low-frequency N supply only when N amount is lower, but the response of *Alternanthera philoxeroides* to N frequency was not related to N amount (Wang et al., 2015). To our knowledge, the effect of N pulse has rarely been investigated at the functional group scale (James and Richards, 2005; Jankju-Borzelabad and Griffiths, 2006; Osone et al., 2014).

We selected 12 common grassland species of three functional groups (i.e., grasses, forbs and legumes) to investigate effects of N frequency and amount on biomass production and root performance at both species and functional group scales. Additionally, we examined the relationship between root responses to N frequency and root size and between biomass production and root morphology. We proposed the following specific hypotheses: (1) different amounts of N supply trigger contrasting root morphology and biomass allocation, and high-amount N supply enhances plant performance; (2) high-frequency N supply is more beneficial for plant performance than low-frequency supply, and such an effect of N frequency depends on N amount; (3) the effects of N amount and N frequency vary at both species and functional group scales; (4) the magnitude of root responses to N frequency depends upon the root system size in the term of final root mass.

2. Materials and methods

2.1. Experimental design

The 12 species (4 legumes, 4 grasses and 4 forbs) are common grassland species native to and also widespread in China. They produce roots over a wide range of size. The seeds of these species were all provided by Golden Earth Agricultural Technology Research Institute, Beijing (website: http://www.jtdseed.com). More detailed descriptions of each species are shown in Table 1, and available at http://www.floraofchina.org.

The experiment took a factorial design, with two levels of N amount (low amount *vs*. high amount) crossed with two levels of N pulse frequency (low-frequency *vs*. high-frequency), resulting in a total of four treatments. There were five replicates for each treatment. Low-amount N supply was set to 2.59 g m^{-2} yr^{-1}, presenting the average level of current N deposition in the eastern China, and high-amount N supply was 10.36 g m^{-2} yr^{-1}, reflecting a possible level of future N deposition (Lu and Tian, 2007).

On June 1, 2014, 250 seeds of each species were sown in 50-cell trays filled with a 1:1 (v:v) mixture of sand and peat (Pindstrup Seedling; Pindstrup Mosebrug A/S, Pindstrup, Denmark). The peat was specially designed for cultivation of seedlings, containing both macronutrients (4.341 mg N L^{-1}, 2.138 mg P L^{-1}, 7.128 mg K L^{-1} and 0.518 Mg mg L^{-1}) and micronutrients (data from the instruction of "Pindstrup Seedling" peat). Total N in the sand-peat mixture was 5.74 ± 1.56 mg N g^{-1} (mean ± SE, $N = 4$). After one month of growth, 20 similar-sized seedlings (approx. 15 cm tall) of each species were randomly selected, and there were 240 plants in total for the 12 species. These plants were transplanted into 240 pots (14 cm in diameter × 12 cm in height) filled with the sand-peat mixture, with one plant per pot. N was supplied in the form of the NH_4NO_3 solution. In the low-amount treatment each pot was supplied with a total of 12 mg N during the experiment, and in the high-amount treatment each pot was supplied with a total of 48 mg N. In the low-frequency treatment, each pot was supplied with N (0.75 g for the low-amount treatment and 3 g for the high-amount treatment) for 16 times (once every three days) during the 48-day experiment, and in the high-frequency treatment it was supplied with N (6 g for the low-amount treatment and 24 g for the high-amount treatment) for only twice (once every 24 days).

The experiment was conducted over a period of 48 days in a greenhouse at Forest Science Company of Beijing Forestry University in Beijing, China. During the experiment, daily mean temperature was 29.44 °C (SE: 0.20 °C) and mean humidity was 64.23% (SE: 0.63%), as measured by iButtons (model DS1923; Maxim Integrated Products, Sunnyvale, CA, USA). During the experiment, the day length in Beijing was about 14 h per day (http://www.wunderground.com).

2.2. Measurements and data analyses

At the end of the experiment, all plants were harvested and the whole root length of each plant was measured using the WinRHIZO Pro 2004a software (Regent Instruments, Inc., Québec, Canada). The plants were then divided into leaves, stems and roots, oven-dried at 70 °C for 48 h, and weighed. Before analysis, we calculated total mass (i.e., the sum of leaves, stems and roots), root mass, root length, root mass fraction (i.e., root mass/total mass; RMF), and specific root length (i.e., root length/root mass; SRL). Total mass and root mass were natural log-transformed to meet the assumption of homogeneity of variance. Root biomass was also used as a measure of the size of the root system (Campbell et al., 1991; Wijesinghe et al., 2001).

We used three-way nested ANOVAs to test the effects of functional group (fixed factor), N amount (fixed factor), N frequency (fixed factor), and species identity (nested within the functional group factor) on total mass, root mass, root length, RMF, and SRL. Linear contrasts were conducted to determine the differences between treatments.

Regressions were employed to examine the relationships between biomass and root related measures within each of the four N treatments. The log response ratio of root mass was calculated to measure the relative response strength of each species to the N frequency treatment within each N amount treatment, i.e., LnRR = Ln (value in the low-frequency treatment/value in the high-frequency treatment). The distribution of LnRR is symmetrical around zero, with negative values indicating that the plant benefits more from the high-frequency treatment and positive values indicating the plant benefits more from the low-frequency treatment. One-sample *t*-tests were used to examine whether LnRR differed significantly from zero. All analyses were conducted using SPSS 22.0 (SPSS, Inc., Chicago, IL, USA).

Table 1
List of species used in the experiment.

Species	Family	Life history	Func. group	Abbrev.
Agropyron cristatum (L.) Gaertn.	Poaceae	Perennial	Grass	Ac
Cynodon dactylon (L.) Pers.	Poaceae	Perennial	Grass	Cd
Lolium perenne L.	Poaceae	Perennial	Grass	Lp
Setaria viridis (L.) P. Beauv.	Poaceae	Annual	Grass	Sv
Amaranthus hypochondriacus L.	Amaranthaceae	Annual	Forb	Ah
Cichorium intybus L.	Asteraceae	Perennial	Forb	Ci
Plantago asiatica L.	Plantaginaceae	Perennial	Forb	Pa
Silphium perfoliatum L.	Asteraceae	Perennial	Forb	Sp
Astragalus adsurgens Pall.	Fabaceae	Perennial	Legume	Aa
Astragalus sinicus L.	Fabaceae	Annual	Legume	As
Melilotus officinalis (L.) Pall.	Fabaceae	Biennial	Legume	Mo
Medicago sativa L.	Fabaceae	Perennial	Legume	Ms

Nomenclature and authorities follow the *Flora of China.*

3. Results

3.1. N effects at the functional group level

A significant effect of N amount was detected in all growth and morphological measures (Table 2). As N amount increased, biomass, root mass and root length increased, whereas RMF and SRL decreased (Fig. 1). The effect of N pulse frequency strongly depended upon N amount and functional group (i.e., the significant effects of FG × A × F on biomass, root mass and root length in Table 2). Compared to the low-frequency treatment, the high-frequency treatment significantly increased biomass and root growth (root mass and root length) of grasses and forbs when the N amount was high, whereas the high-frequency treatment did not influence biomass and even tended to decrease root growth when the N amount was low (Fig. 1a–c). Interestingly, N frequency also affected SRL of forbs, i.e., the high-frequency treatment tended to increase SRL of forbs when the N amount was low (Fig. 1e).

Table 2
Nested ANOVA results for effects of functional group (FG, fixed), nitrogen amount (A, fixed), pulse frequency (F, fixed), and species identity (S, nested within FG effect) on biomass, root mass, root length, root mass fraction (RMF), and specific root length (SRL) of the 12 plant species.

	df	Biomass	Root mass	Root length	RMF	SRL
Func. Group (FG)	2,9	1.2	0.4	1.1	0.8	0.1
Amount (A)	1,9	**66.4*****	**20.8** **	**6.1***	**12.1****	**9.1***
Frequency (F)	1,9	0.7	0.3	3.7$^{\#}$	0.01	3.2
Species (S)	9,2	**52.8****	**38.8*****	**5.1****	**70.6***	**10.5****
FG × A	2,9	2.4	1.4	1.9	0.1	0.3
FG × F	2,9	2.3	0.6	1.0	3.1 $^{\#}$	2.9
A × F	1,9	**5.5***	8.2	**6.5***	0.1	4.0$^{\#}$
FG × A × F	2,9	**4.9***	**22.7*****	**7.0***	0.9	2.6
A × S	9,9	1.1	**4.4***	**5.3***	1.3	0.5
F × S	9,9	0.8	**3.2***	**3.3***	0.5	1.2
A × F × S	9,192	0.8	0.2	0.4	0.9	1.0

Degrees of freedom (*df*), *F* values and significances are given. (no symbol) $P > 0.1$; $^{\#}P < 0.1$; $^{*}P < 0.05$; $^{**}P < 0.01$; $^{***}P < 0.001$. Values for which $P < 0.05$ are in bold.

3.2. Relationships between biomass and root measures

Biomass was significantly positively correlated with root mass, irrespective of N frequency or amount treatments, and negatively correlated with SRL in three of the four N treatments, the exception being the high-frequency and high-amount treatment (Fig. 2a and d). No significant correlation was detected between biomass and root length or between biomass and RMF in any of the N treatments (Fig. 2b and c).

3.3. Responses to N frequency and root size

The magnitude of the response to N frequency as measured by LnRR was significantly negatively correlated with root size of the plants when the N amount was high (Fig. 3). The values of LnRR for the three species with larger root systems (*Cynodon dactylon*, *Lolium perenne* and *Plantago asiatica*; Fig. 3a) were significantly below zero, whereas the values in *Cichorium intybus* and *Silphium perfoliatum* tended to be negative, and the values for other species with smaller root systems, including the three legume species, were close to zero (Fig. 3a). Conversely, no negative relationship was observed when the N amount was low (Fig. 3b). In the low-amount treatment, the values of LnRR for the three legume species with smaller root systems (*Astragalus adsurgens*, *Astragalus sinicus* and *Melilotus officinalis*) were significantly (or tended to be) below zero, whereas the values in the non-legume species with smaller root systems (*Agropyron cristatum*, *Amaranthus hypochondriacus*, and *C. intybus*) were above zero (Fig. 3b).

4. Discussion

4.1. Plant responses to N supply amount

A positive effect of the increased N supply amount on plant performance has been reported in many previous studies (Fransen et al., 1998; Li et al., 2011; Xia and Wan, 2008), and is clearly exhibited in the present work. As the amount of N supply increased, biomass increased by 78.1%–210.9% and root mass increased by 45.6%–101.5% in the majority of grasses and forbs, although neither biomass production nor root growth was changed in legumes. Furthermore, the majority of grasses or forbs adjusted root growth, morphology and biomass allocation in response to different N-amount treatments, e.g., the high-amount N supply stimulated an increase in the thickness and length of roots, and a decrease in root mass investment. In addition, there was a positive correlation between biomass and root mass and a negative correlation between biomass and SRL (Fig. 2). These results imply that root growth and SRL strongly influenced plant performance (Suding et al., 2005; Xia and Wan, 2008), and that the effect of N amount is dependent on

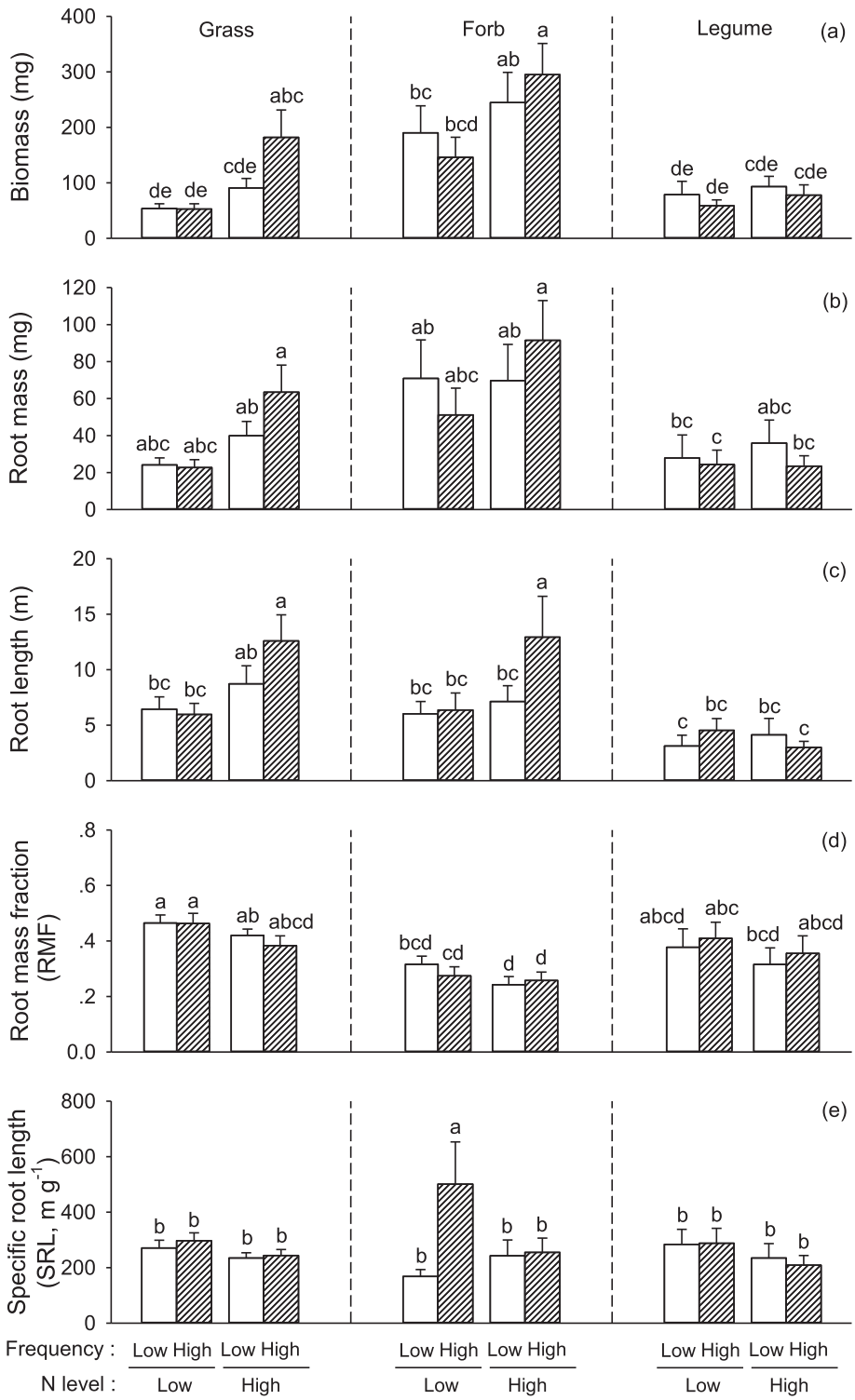

Fig. 1. Effects of functional group, nitrogen amount, and pulse frequency on biomass, root growth and morphological measures of the three plant functional groups. Means with same letter are not statistically different (Linear contrasts, $P = 0.05$).

functional groups (Gough et al., 2012; Pennings et al., 2005; Xu et al., 2015). The change in biomass allocation under different N-amount treatments was consistent with the classic theory of biomass allocation, i.e., an increase in the proportional biomass allocated to produce the structures necessary for acquiring the limiting resources (Bloom et al., 1985; Thornley, 1972). Overall, the results provide support for our first hypothesis stating that a high N supply can trigger root morphological plasticity and biomass allocation, and positively affect plant performance.

4.2. Plant responses to N pulse frequency

Compared to the low-frequency treatment, the high-frequency treatment had a positive effect on the biomass production or root

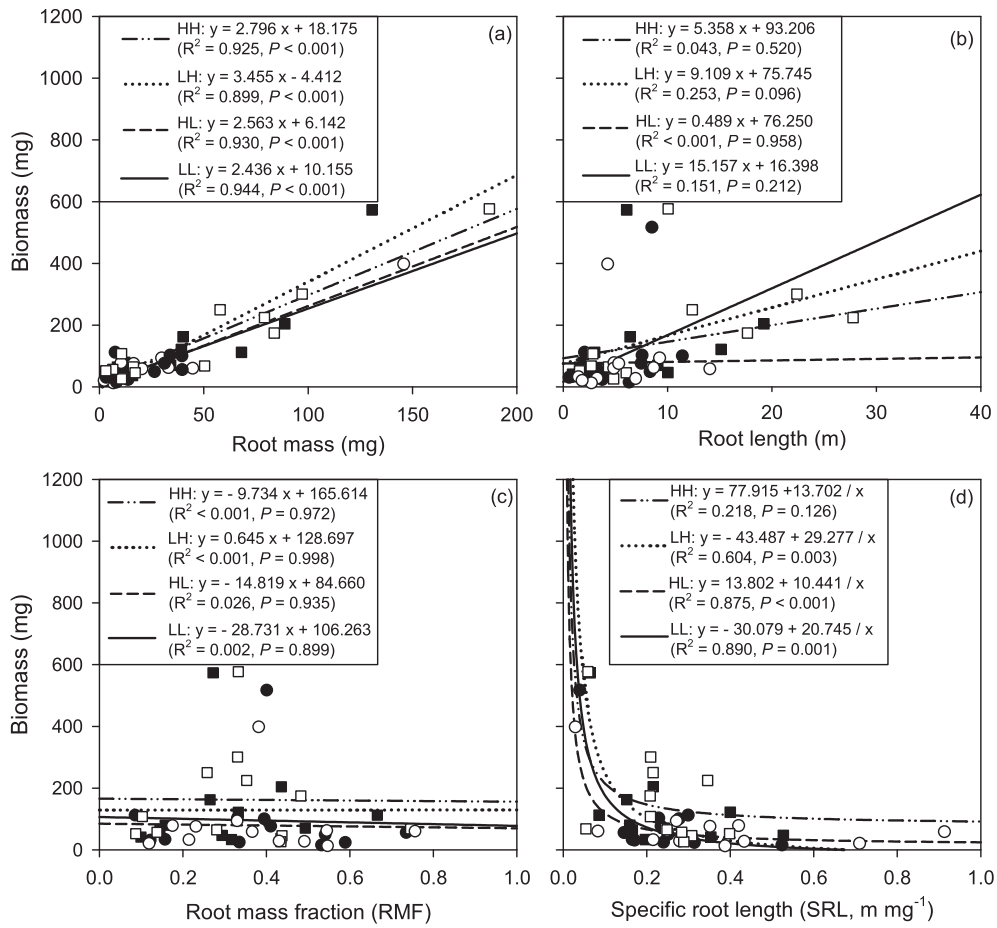

Fig. 2. The relationships between root traits and final biomass within the 12 plant species in each N treatment. Symbols indicate different N treatments; □ High frequency - high amount treatment (HH); ■ Low frequency - high amount treatment (LH); ○ High frequency - low amount treatment (HL); ● Low frequency - low amount treatment (LL).

growth of non-legume species such as *A. cristatum*, *C. dactylon*, *L. perenne*, and *P. asiatica* (unpublished data). The N frequency effect on root morphology (as measured by specific root length) was only detected in forbs. The results provide support for the second hypothesis that high-frequency supply is more beneficial for plant growth compared to low-frequency supply, and the third hypothesis that the N frequency effect varies at both plant species and functional group levels (Craine et al., 2003; Lamb et al., 2012). One plausible reason for these observations is the difference in root structure and function among species. In particular, the symbiosis between legume species and rhizobia might to some degree simplify the inherent root architecture and/or function of legume species (Hodge et al., 2009; Goh et al., 2015), thereby limiting the adaptive morphological plasticity of legumes in response to N supply. This is because the inoculation of rhizobia could stimulate root development and plasticity of host plants by altering their root morphology (e.g., total root length and surface areas) and mass (Souleimanov et al., 2002), and also by triggering later root formation (Oláh et al., 2005). The other plausible reason is the difference in root size among species. The final root size (i.e., mass) in the majority of non-legume species (i.e., 8.44 mg root mass in grass species and 8.11 mg in forb species) was 2.24 times larger than that in legume species (i.e., 3.70 mg). Compared to small root systems, larger root systems might provide non-legume species with enhanced sensitivity to fluctuations in N amount caused by pulsed N supply, and to maintain a higher root uptake efficiency, although larger plant size unavoidably causes higher metabolic costs (Lamb et al., 2012).

4.3. Relationships between root responses to N pulse and root size

The current experiment covering multiple species allowed us to examine the relationship between the magnitude of the response to pulsed N supply and root system size under two N availability levels. As predicted by the fourth hypothesis, there was an observed negative correlation between root response to N frequency (as indicated by the log response ratio of root mass) and root size, i.e., species producing larger roots were more sensitive to a high-frequency supply of N than to a low-frequency supply. The magnitude of the response would gradually approach zero as root size decreased, which signified no obvious difference between the two N frequency treatments. However, such a relationship only occurred when the amount of N supply was high, and did not occur when the amount of N was low. The results indicate that the response to N frequency is an emergent property of plants when confronted with low N supply. These results are consistent with previous work, suggesting that the response to N frequency may also be modified by the availability of other resources such as water and phosphorus (Drenovsky and Richards, 2004; James and Richards, 2006; Navarro-Angulo and Robledo, 1999). The

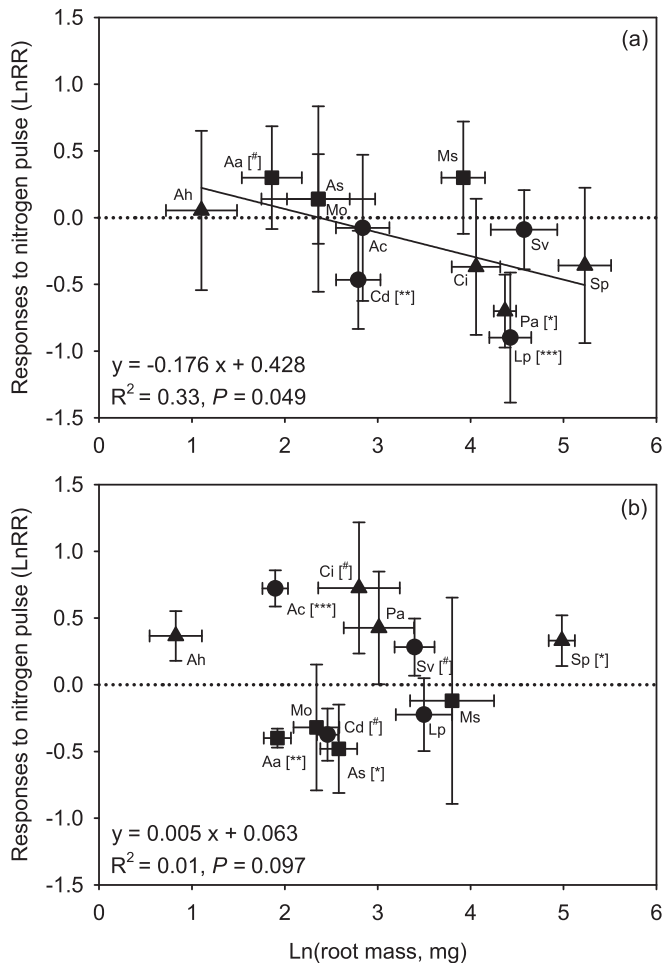

Fig. 3. The relationship between the response to N pulse frequency (LnRR) and root size [Ln (root mass)] with either the high - (a) or low-amount N treatment (b). The three different shaped points represent the three plant functional groups, i.e., grasses (circle), forbs (triangle), and legumes (square). Species abbreviations correspond to those listed in Table 1. Different symbols in "[ ]" indicate the significances between the value of LnRR and zero; (no symbol) $P > 0.1$; $^{\#}P < 0.1$; $^{*}P < 0.05$; $^{**}P < 0.01$; $^{***}P < 0.001$.

relationship also suggests that it is possible to predict the growth response of plants to pulsed supply of N based on root size of individuals, and on the premise that the total amount of N supply is high. In addition, root response to N pulse (as indicated by log response ratio) of the three annual species was all close to zero in the high amount N treatment, while that of most perennials was relatively far away from zero. This result implies that perennials might be better adapted to N pulse than annuals.

5. Conclusions

Non-legume species appear to perform better under high-frequency N supply than under low-frequency supply, whereas legume species cannot acclimatize to pulsed N supply. The contrasting responses to N supply might ultimately result in different habitat preferences of species, where non-legume species with larger root systems (e.g., *Cynodon dactylon*, *Lolium perenne* and *Plantago asiatica*) prefer to colonize fertilized habitats with high-frequency (or continuing) N supply, and legume species have greater competitive advantages in unfertilized habitats. The relationship between the magnitude of root response and root size also provides the possibility to predicate the growth pattern of species under pulsed N supply, at least when the total amount of N supply is high. One limitation of our study is that the experiment duration was relative short (48 days) so that the long-term effect of treatments could not be assessed. While the soil mixture that we used might contain rhizobia, we do not know whether the quantity was sufficient for the growth of the legumes. Furfure studies may be conducted to address whether root responses to manipulated N reflect the success of plants under natural conditions.

Author contributions

BCD and FHY conceived and designed the experiment. QYY, PW and LL conducted the experiment. BCD and QYY analyzed the data. QYY, BCD and FHY wrote the manuscript; other authors provided editorial advice. All authors have approved the final version of the article.

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